



Training Piscine datascience - 4

The futur

Summary: Today, you will see some model to predict the future

Version: 1.00

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Chapter I

General rules

- You have to render your modules from a computer in the cluster either using a virtual machine:
 - You can choose the operating system to use for your virtual machine
 - Your virtual machine must have all the necessary software to realize your project. This software must be configured and installed.
- Or you can use the computer directly in case the tools are available.
 - Make sure you have the space on your session to install what you need for all the modules (use the goinfre if your campus has it)
 - You must have everything installed before the evaluations
- Your functions should not quit unexpectedly (segmentation fault, bus error, double free, etc) apart from undefined behaviors. If this happens, your project will be considered non functional and will receive a 0 during the evaluation.
- We encourage you to create test programs for your project even though this work **won't have to be submitted and won't be graded**. It will give you a chance to easily test your work and your peers' work. You will find those tests especially useful during your defence. Indeed, during defence, you are free to use your tests and/or the tests of the peer you are evaluating.
- Submit your work to your assigned git repository. Only the work in the git repository will be graded. If Deepthought is assigned to grade your work, it will be done after your peer-evaluations. If an error happens in any section of your work during Deepthought's grading, the evaluation will stop.
- By Odin, by Thor ! Use your brain !!!

Chapter II

Introductions

Data Scientists often use tools such as python, Jupyter Notebook, Julia ...

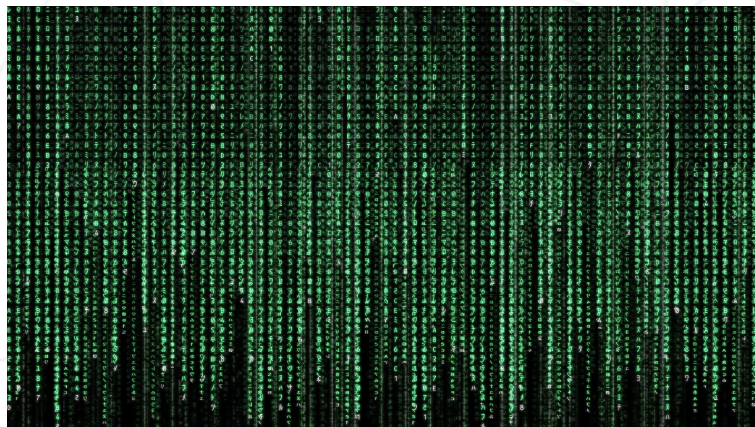
It is up to you to find the tools that suit you. You are free to use any language of your choice for this module.

The role of the data scientist is to predict "the future" with automatic learning models on past data, he must be a force of proposal to explain the possible interest to the implementation of his models, create tools to help decision making.

Chapter III

Exercise 00

	Exercise 00
Exercise 00 : Confusion Matrix	
Turn-in directory : <i>ex00/</i>	
Files to turn in : Confusion_Matrix.*	
Allowed functions : lib to display the image only	

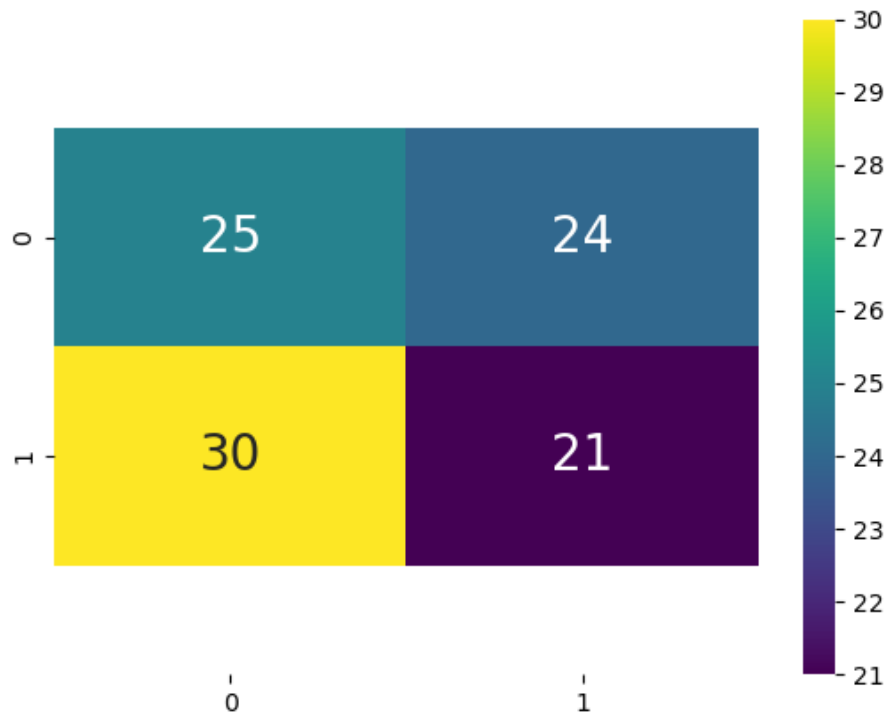


- The data seems too Multicollinearity, you will have to make a Detecting Multicollinearity model
- Print and display matrix confusion

```
$>./Confusion\_Matrix.* predictions.csv truth.csv
precision recall f1-score total
0      0.51    0.45    0.48    55
1      0.41    0.47    0.44    45

accuracy          0.46    100

[[25 24]
 [30 21]]
$>
```



Make sure that this exercise work with both strings and numbers. If this exercise is wrong, the eval stop here : it's over.



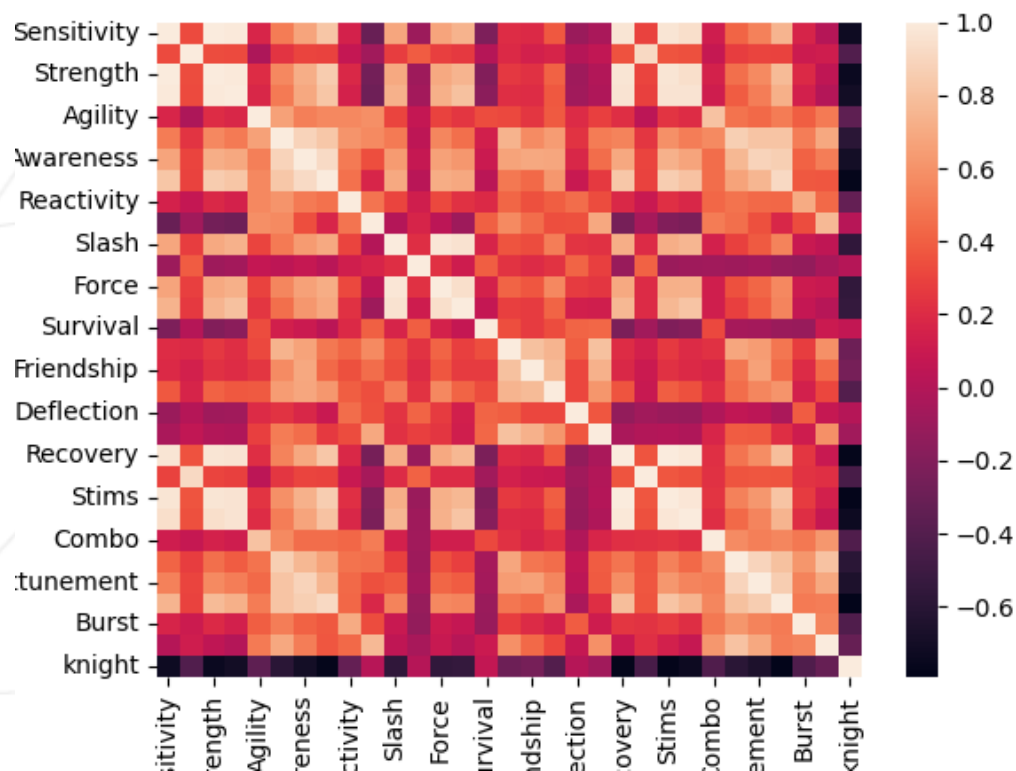
Make sure you understand this exercise as it will be checked during the evaluation

Chapter IV

Exercise 01

	Exercise 01
Exercise 01 : It is warm	
Turn-in directory : <i>ex01/</i>	
Files to turn in : Heatmap.*	
Allowed functions : All	

- Make a Heatmap to see the Correlation Coefficient between the data

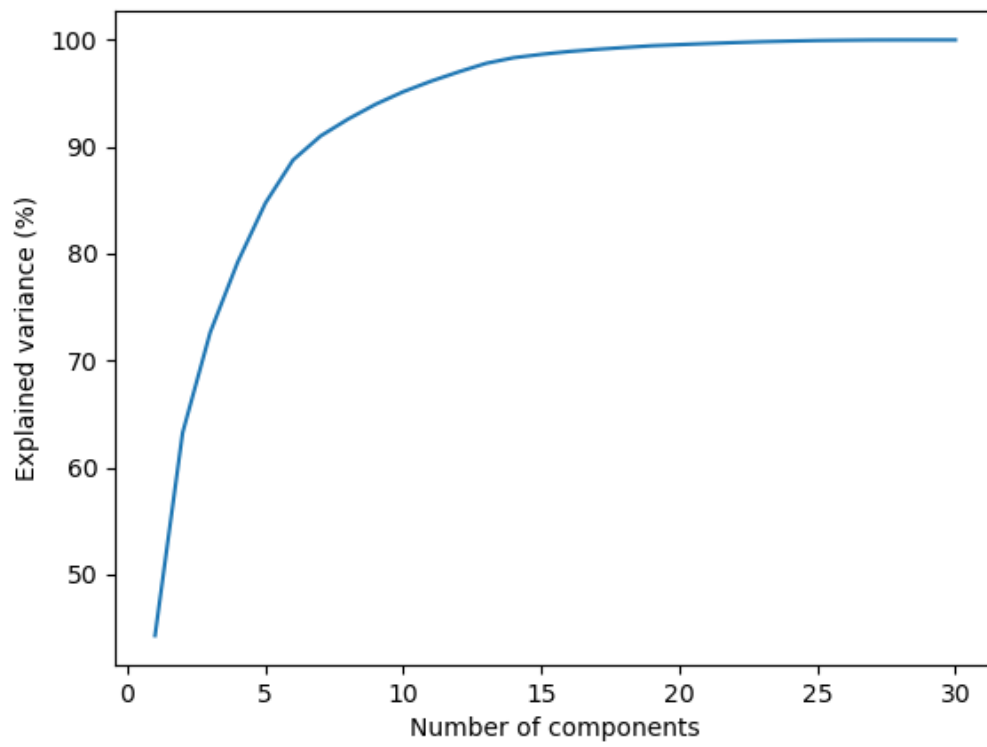


Chapter V

Exercise 02

	Exercise 02
Exercise 02 : Variances	
Turn-in directory : <i>ex02/</i>	
Files to turn in : variances.*	
Allowed functions : A11	

- Add up the variances to see how many components are useful
- Display a graph representing the curve



Chapter VI

Exercise 03

	Exercise 03
Exercise 03 : Feature Selection	
Turn-in directory : <i>ex03/</i>	
Files to turn in : Feature_Selection.*	
Allowed functions : All	

The data seems too Multicollinearity, you will have to make a Detecting Multicollinearity model.

There are a number of models for detecting the importance of variables (Lasso, Backward Elimination, Step Forward Selection, ...)

But here you will have to use the Variance Inflation Factor (VIF)

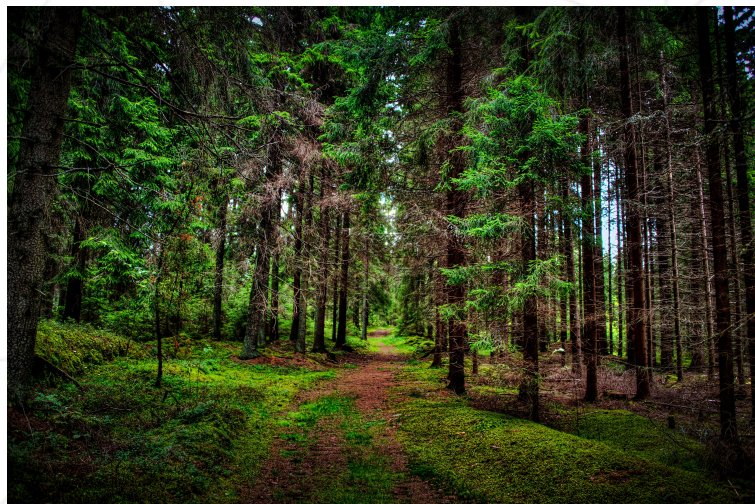
- Display the VIF of your data
- Keep only the features so that the VIF goes under 5, and display the features

	VIF	Tolerance
Empowered	36.763714	0.027201
Stims	405.023336	0.002469
Prescience	60.041733	0.016655
Recovery	799.105946	0.001251
Strength	3786.400419	0.000264
...	...	...

Chapter VII

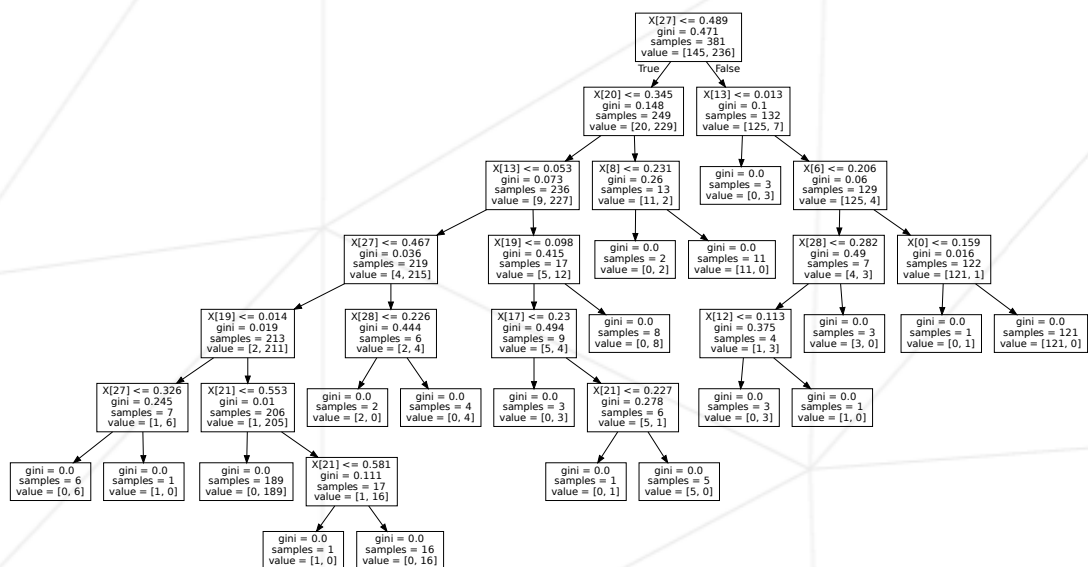
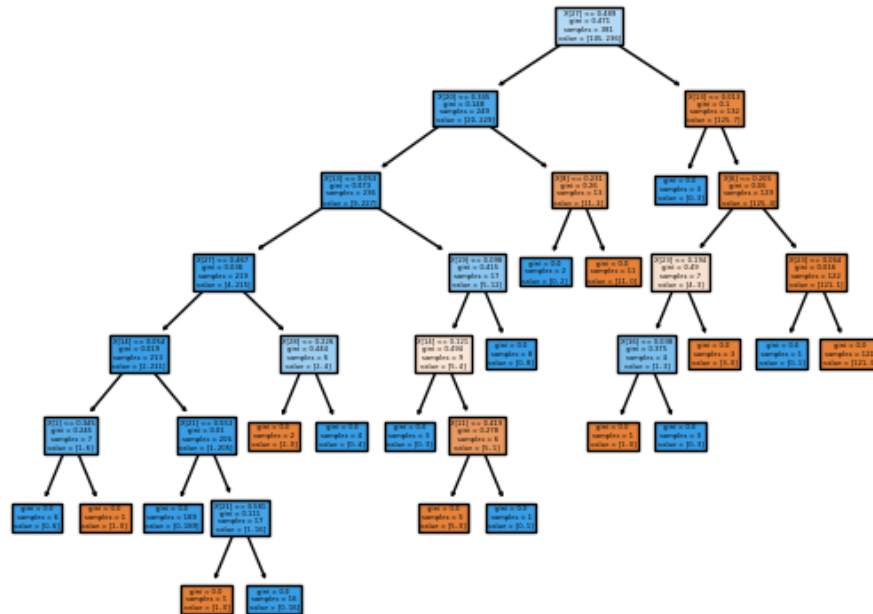
Exercise 04

	Exercise 04
Exercise 04 : Forest	
Turn-in directory : <i>ex04/</i>	
Files to turn in : Tree.*	
Allowed functions : All	



- Make a Decision Tree Classifier or Random Forest Classifier model
- Display the tree in a graph
- Your program must take the `Train_knight.csv` file as first argument, the `Test_knight.csv` (provided during the evaluation) file as second argument and write a `Tree.csv` file with the prediction (one prediction per line, either `jedi` or `sith`)
- Your model must have minimum 90% of f1-score

Decision tree trained on all Knights features



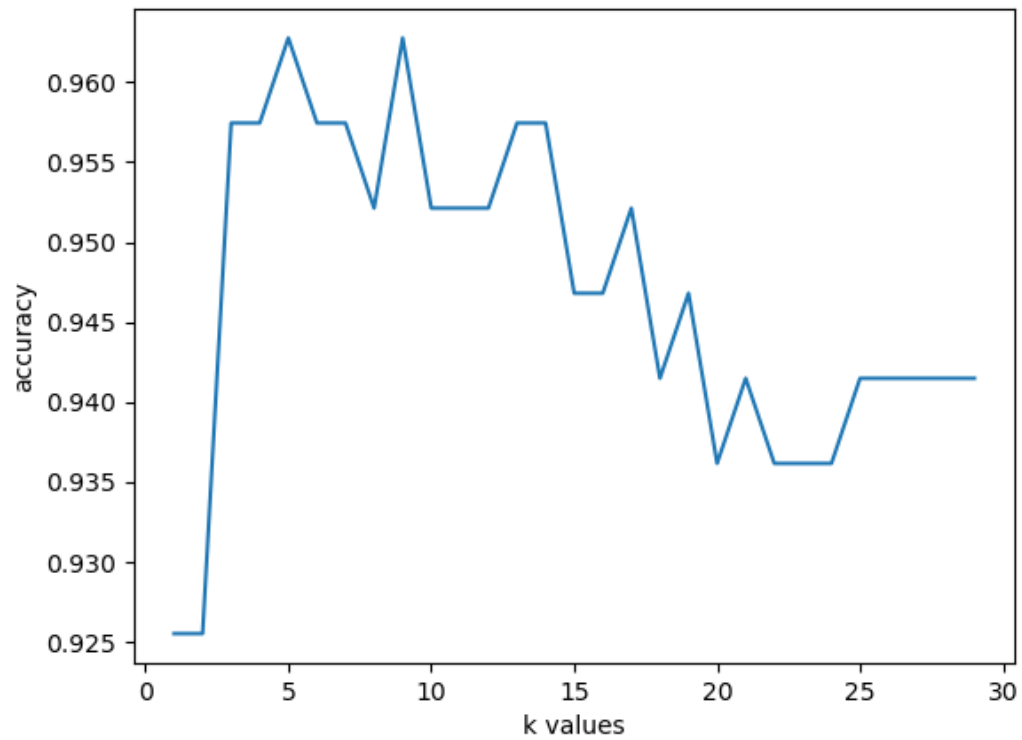
Create your Test_knight.csv file for your tests

Chapter VIII

Exercise 05

	Exercise 05
Exercice 05 : KNN	
Turn-in directory : <i>ex05/</i>	
Files to turn in : <i>KNN.*</i>	
Allowed functions : <i>A11</i>	

- Your program must take the *Train_knight.csv* file as first argument, the *Test_knight.csv* (provided during the evaluation) file as second argument and write a *KNN.csv* file with the prediction (one prediction per line, either *jedi* or *sith*)
- Make a KNN that calculates the precision % according to the number of k-value
- Display the graph
- You must have a minimum of 95% f1-score



Create your Test_knight.csv file for your tests



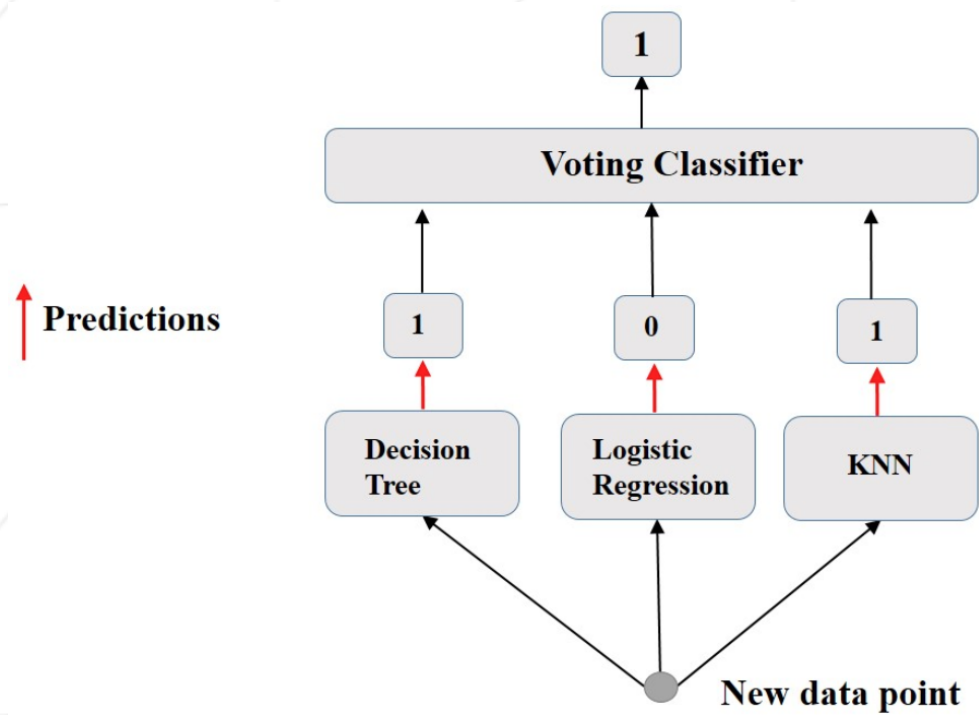
Do you have convergence problems?

Chapter IX

Exercise 06

	Exercise 06
Exercise 06 : democracy !	
Turn-in directory : <i>ex06/</i>	
Files to turn in : democracy.*	
Allowed functions : A11	

- Choose a third model of your choice
- Make a Voting classifier
- Your program must take the Train_knight.csv file as first argument, the Test_knight.csv (provided during the evaluation)file as second argument and write a Voting.csv file with the prediction (one prediction per line, either jedi or sith)
- You must have a minimum of 96% f1-score



Create your Test_knight.csv file for your tests

Chapter X

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double check the names of your folders and files to ensure they are correct.



The evaluation process will happen on the computer of the evaluated group.